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Curriculum Vitae – Gravitational Wave Astronomy and Numerical Relativity

Publications

Published and archived

19. **Vaishak Prasad**, *Prayush Kumar, Michael Boyle, Lawrence E. Kidder, Harald P. Pfeiffer, Mark A. Scheel*, "A catalog of long, eccentric and precessing binary black hole simulations from ICTS-TIFR", ArXived 14 October 2026, gr-qc:2610.19688.
Submitted.

18. Yu-Cun Xie, ***Vaishak Prasad***, "Universal Structure of Horizon Formation in Generic Binary Black Hole Mergers", ArXived 10 September 2026, gr-qc:2609.11819.

17. **Vaishak Prasad**, "NRHJSur3dq8: a spectral numerical-relativity surrogate family for non-eccentric aligned-spin binary-black-hole waveforms", ArXived 8 September 2026, gr-qc:2609.09088.
Single author. Released with the public `nrhjsurrogate` package.

16. **Vaishak Prasad**, *Monica Rincón-Ramírez, Nathan K. Johnson-McDaniel, Eugenio Bianchi, Ish Gupta and B. S. Sathyaprakash*, "Do black hole mergers maximize entropy?", Published 27 August 2026, Int. J. Mod. Phys. D, 2641013..
Gravity Research Foundation 2026 essay, Honourable Mention.

15. Monica Rincón-Ramírez, *Nathan K. Johnson-McDaniel, Eugenio Bianchi, Ish Gupta,* ***Vaishak Prasad*** *and B. S. Sathyaprakash*, "Maximum Entropy Conjecture for Black Hole Mergers", Published 10 July 2026, Phys. Rev. Lett. 137, 021406., arXiv:2601.22388.
Devised the numerical relativity route used to test the conjecture, framed in Bondi mass and angular momentum, and contributed the horizon-geometry analysis.

14. **Vaishak Prasad**, "Accelerated Time-domain Analysis for Gravitational Wave Astronomy", ArXived 6 March 2026, gr-qc:2603.06010.
Single author. Exact time-domain likelihood evaluation at near-FFT cost via the Gohberg–Semencul representation. Released with the `tdanalysis` package.

13. **Vaishak Prasad**, "Multi-Segment Consistency Tests of General Relativity", ArXived 6 March 2026, gr-qc:2603.05835.
Single author. Demonstrated on GW250114; to be proposed as a reviewed LVK test of general relativity.

12. **Vaishak Prasad**, *and B. S. Sathyaprakash*, "Are black hole spins truly near-zero?", ArXived 13 February 2026, gr-qc:2602.12859.

11. Divyajyoti, *Isobel M. Romero-Shaw,* ***Vaishak Prasad****, Kaushik Paul, Chandra Kant Mishra, Prayush Kumar, Akash Maurya, Michael Boyle, Lawrence E. Kidder, Harald P. Pfeiffer, Mark A. Scheel*, "Biased parameter inference of eccentric, spin-precessing binary black holes", Published 12 May 2026, Phys. Rev. D 113, 103022., arXiv:2510.04332.

10. **LVK collaboration**, *incl.* ***Vaishak Prasad***, "Black Hole Spectroscopy and Tests of General Relativity with GW250114", Published 30 January 2026, Phys. Rev. Lett. 136, 041403., arXiv:2509.08099.
Made critical contribution to the analysis and interpretation of the area law result, its connection with consistency test, the ringdown model, and review of results. Editors' Suggestion; featured in *Physics*.

9. **LVK collaboration**, *incl.* ***Vaishak Prasad***, "GW250114: Testing Hawking's Area Law and the Kerr Nature of Black Holes", Published 10 September 2025, Phys. Rev. Lett. 135, 111403..
Made critical contribution to the analysis and interpretation of the area law result, and ringdown model

8. **LVK collaboration**, *incl.* ***Vaishak Prasad***, "GW230814: investigation of a loud gravitational-wave signal observed with a single detector", Published 18 June 2026, Astrophys. J. Lett. 1004, L23., arXiv:2509.07348.
   Contributed the targeted NR simulation and to paper writing.
7. **Vaishak Prasad**, "The Shear at the Common Dynamical Horizon in Binary Black Hole Mergers and its Imprint in their Gravitational Radiation.", Published 28 April 2025, Phys. Rev. D 111, 084070..
6. P. Ajith et. al, *incl.* ***Vaishak Prasad***, "The Lunar Gravitational-wave Antenna: Mission Studies and Science Case.", Published 28 January 2025, JCAP 01 (2025) 108..
   Author of the Testing GR section
5. **Vaishak Prasad**, "Tidal deformation of dynamical horizons in binary black hole mergers and its imprint on gravitational radiation.", Published 16 February 2024, Phys. Rev. D 109, 044033., arXiv:2109.01193.
4. Sashideep Gutti, *Shailesh Kulkarni,* ***Vaishak Prasad***, "Closed timelike curves and energy conditions in regular spacetimes", Published 15 December 2022, Eur. Phys. J. C (2022) 82: 1136..
   Equal contribution
3. **Vaishak Prasad**, *Anshu Gupta, Sukanta Bose, Badri Krishnan*, "Tidal deformation of dynamical horizons in binary black hole mergers.", Published 9 February 2022, Phys. Rev. D 105, 044019., arXiv:2106.02595.
2. **Vaishak Prasad**, *Anshu Gupta, Sukanta Bose, Badri Krishnan, and Erik Schnetter*, "News from Horizons in Binary Black Hole Mergers", Published 16 September 2020, Phys. Rev. Lett. 125, 121101., arXiv:2003.06215.
1. **Vaishak Prasad**, *Rahul Srinivasan, Sashideep Gutti*, "The Chronology Protection Problem in modified Kerr-Newman space-times.", Published 14 January 2019, Phys. Rev. D 99, 024023., arXiv:1803.06666.

Submitted and in preparation

10. **Vaishak Prasad**, *B. S. Sathyaprakash and Abhay Ashtekar*, "GW250114 and the test of the black hole areas"., In preparation.
9. **Vaishak Prasad**, *Abhay Ashtekar and B. S. Sathyaprakash*, "Assessing intrinsic waveform accuracy using the waveform balance laws"., In preparation.
8. **Vaishak Prasad**, "Waveform systematic measurements using the multi-segment consistency test"., LIGO-P2600044-v1, in preparation.
7. **Vaishak Prasad**, "Parallel Template Bank Generation for Non-Eccentric Precessing Binaries"., LIGO-P2600567, in LIGO-Virgo-KAGRA internal review (P&P).
6. **Vaishak Prasad**, *B. S. Sathyaprakash and Abhay Ashtekar*, "A universal law for the balance-law violation of finite-length gravitational waveforms: post-Newtonian origin and the geometry of precession"..
5. Viviana Caceres, ***Vaishak Prasad***, "Time-domain black-hole spectroscopy with a segmented, GPU-accelerated likelihood"..
4. **Vaishak Prasad**, "Cauchy-characteristic matching via generalized-harmonic evolution on compactified null slices. I. Formulation and the hyperbolicity of exact null slicing"..
3. **Vaishak Prasad**, "Batched likelihood evaluation for nested sampling in gravitational-wave parameter estimation"..
2. **Vaishak Prasad**, "Compile-time performance optimization of numerical relativity simulations in C++ using SpEC"..
1. Priyanka Sinha, *Samanwaya Mukherjee, Prayush Kumar,* ***Vaishak Prasad****, Lawrence Kidder, Harald Pfeiffer, Mark Scheel, Md Shaikh, Michael Boyle*, "Can we twist gravitational waveforms from planar eccentric binaries to emulate generic ones"..